

Marwan Abdelaal

Backend Engineer · Full-Stack Developer · AI Engineering Intern

marwan.abdelaal@outlook.com · +201111210005 · Cairo, Egypt · [linkedin.com/in/marwan-abdelaal](#) · [github.com/MoscowAbdelaal](#)

PROFILE

3rd-year Software Engineering student at GIU (GPA: 2.4, Very Good, German Scale) with a Minor in IT Security. I build backend systems that work in the real world: production APIs, durable schedulers, idempotent publishing pipelines, SaaS billing engines, and AI-integrated platforms. Currently interning at FlyRank AI where I shipped two production-grade capstone systems and a suite of backend assignments. Led a team of 10 to ship an AI-powered MERN career platform. Holder of 20 Anthropic Academy certifications in AI fluency and Claude tooling. I take ownership, ship things, and learn fast.

SKILLS

Backend:	Node.js · Express.js · REST APIs · JWT · Middleware · Supabase Auth · Swagger UI · BullMQ
Languages:	JavaScript · Python · Java · C# · SQL · HTML & CSS
Databases:	PostgreSQL · MongoDB · SQLite · SQL Server · Redis · Schema Design
AI & ML:	Hugging Face · Ollama · NLP · Prompt Engineering · Embeddings · Zero-Shot Classification
Security:	AES-GCM · HMAC · 3DES · RSA · Diffie-Hellman · OAuth · Webhook Verification
Reliability:	Idempotency · Durable Scheduling · Quota Enforcement · Rate Limit Handling · Retry Logic
Payments:	Stripe Test Mode · Stripe CLI · Checkout · Subscription Webhooks · Token Pricing
DevOps:	Docker · Docker Compose · Git · GitHub · GitLab · CI/CD
Frontend:	React.js (Vite) · Context API · React Router · Axios
Data:	Cheerio · Zod · pandas · scikit-learn · Jupyter Notebook

EXPERIENCE

- Backend AI Engineering Intern** **FlyRank AI**

Jun 2026 - Present | Remote

 - Shipped two production-grade capstone systems: a multi-platform social campaign publisher (idempotency, BullMQ/Redis scheduling, HMAC webhooks, AES-GCM token encryption) and a SaaS usage metering and billing engine (exactly-once metering, quota enforcement, AI token pricing, Stripe test mode integration).
 - Built a production REST API with 12 endpoints, 23 automated tests at 100% pass rate, JWT auth via Supabase, and a clean PostgreSQL/Docker repository-pattern migration.
 - Developed a web scraping pipeline with Zod validation, exponential backoff, caching reducing runtime from 45s to 1s, and Ollama local AI enrichment.
 - Completed 20 Anthropic Academy certifications covering AI fluency, Claude API, MCP, agent skills, Amazon Bedrock, and Google Cloud.
- Part-Time Teaching Assistant** **ICTeam El Komy - IGCSE ICT**

Jun 2024 - May 2026 | Cairo, Egypt

 - Guided 20+ students through technical concepts and lab sessions, troubleshooting issues in real time and adapting content delivery based on feedback.
- Event Volunteer & Usher** **Cairo International Film Festival & GIU Events**

2024 - Present | Cairo, Egypt

 - Coordinated operations across 3+ large-scale public events, managing cross-functional volunteer teams under high-pressure conditions.

PROJECTS

Usage Metering & Billing Engine - FlyRank Capstone

Node.js · Express.js · PostgreSQL · Docker · Stripe Test Mode · Stripe CLI | [github.com/MoscowAbdelaal/flyrank-capstone-metering-billing](#)

- SaaS billing engine with exactly-once metering (idempotency keys prevent double-charging), quota enforcement with honest 429/402 boundaries, and AI token cost calculation handling cached input and reasoning token pricing separately.
- Stripe test mode integration with HMAC signature-verified webhooks rejecting forgeries with 400, deduplicating replayed events, and syncing tenant plan status only on verified delivery.

Multi-Platform Social Campaign Publisher - FlyRank Capstone

Node.js · Express.js · BullMQ · Redis · PostgreSQL · Docker · Sharp · AES-GCM · HMAC | github.com/MoscowAbdelaal/flyrank-capstone-social-studio

- Converts a blog post into a full social campaign with platform-correct image variants (Instagram 1080x1080, X 1600x900) and platform-tailored captions from reusable prompt fragments.
- Reliability core: idempotent publishing, 429/Retry-After backoff, crash-safe BullMQ worker, HMAC-verified webhooks, and AES-GCM encrypted OAuth tokens with random IV.

GIU Nexus - AI-Powered Career & Talent Platform

React (Vite) · Node.js · Express.js · MongoDB · Hugging Face API · JWT · GitLab | github.com/MoscowAbdelaal/GIU-Nexus

- Led a 10-person team to build a full-stack MERN platform with Hugging Face NLP for skill extraction and zero-shot job classification, cosine similarity recommendations, and role-based dashboards for job seekers, recruiters, and admins.

Production CRUD API - Auth, Testing & Docker Migration

Node.js · Express.js · SQLite · PostgreSQL · Docker · Supabase Auth · JWT · Swagger UI | github.com/MoscowAbdelaal/crud-api

- 12-endpoint REST API with 23 automated tests at 100% pass rate, JWT auth, parameterized queries, Swagger docs, and a clean repository-pattern migration from SQLite to PostgreSQL in Docker.

Human Resource Management System (HRMS)

Microsoft SQL Server · Node.js · HTML · CSS · SQL Stored Procedures

- Full EERD and relational schema design, optimized SQL stored procedures as the backend layer, and a multi-role enterprise system covering payroll, attendance, leave, hiring, and performance reviews.

Cryptography Suite - 3DES & Diffie-Hellman/RSA

Python · Jupyter Notebook

- Triple DES from scratch using EDE structure across 16-round blocks, and a secure communication system combining Diffie-Hellman key exchange with RSA digital signatures simulating HTTPS-like secure transmission.

Clash of Clans: Heroes - 2-Player Strategy Game

Java · JavaFX · Eclipse IDE · OOP · CSV Data Loading | github.com/MoscowAbdelaal/Clash-of-clans-Legends

- Complete 2-player turn-based strategy game across 3 milestones with full OOP architecture, custom exception handling, and a JavaFX frontend with real-time state updates and victory detection.

CERTIFICATIONS - ANTHROPIC ACADEMY (2026)

- AI Fluency: Framework & Foundations	- Claude 101
- Claude Code 101	- Building with the Claude API
- Introduction to Model Context Protocol	- Model Context Protocol: Advanced Topics
- Claude with Amazon Bedrock	- Claude on Google Cloud

20 certifications total - full list available at [linkedin.com/in/marwan-abdelaal](https://www.linkedin.com/in/marwan-abdelaal)

EDUCATION

Bachelor of Software Engineering - Informatics & Computer Science Sep 2024 - Present | Cairo, Egypt
German International University (GIU)

GPA: 2.4 / Very Good (German Scale) · Minor: IT Security · Expected Graduation: 2028 · Coursework: DSA, OOP, Database Systems, Data Science, Information Security, Distributed Systems, Media Informatics

LANGUAGES

- Arabic Native	- English Fluent
- German Beginner	- French Beginner